SAP-C02 Comparison Tables

The services that look alike on the exam, side by side. Green = yes · red = no · yellow = partly / retired. The line under each table is the one rule to remember.

1. VM migration options

Options	Creates EC2 instance / AMI	Agent	Continuous replication	Test, cutover or failover	Use for
VM Import/Export	Yes (import image as AMI)	No (upload the image file)	No (one-time)	No	One-off import of a VM image
Server Migration Service (SMS)	Retired → use MGN	—	—	—	Appears in older exam questions only
Application Migration Service (MGN)	Yes	Yes (agentless option for vCenter)	Yes (block-level)	Partly (test and cutover launches)	Rehost migration with a short cutover
Elastic Disaster Recovery (DRS)	Yes	Yes	Yes (block-level)	Yes (drills, recovery, failback)	Disaster recovery for servers

Rule: Moving servers once → MGN. Keeping a ready copy to fail over to (and back) → DRS.

2. Moving data into AWS

Service	Online / offline	Ongoing or one-time	On-premises keeps access (protocol)	Lands in	Best for
DataSync	Online (agent)	Scheduled, incremental, verified	No	S3, EFS, FSx	Bulk or scheduled file/object copy
S3 File Gateway	Online	Ongoing	Yes (NFS / SMB)	S3	Hybrid file shares, backups to S3
Volume Gateway	Online	Ongoing	Yes (iSCSI block)	S3-backed volumes, EBS snapshots	Block storage for on-premises servers
Tape Gateway	Online	Ongoing	Yes (virtual tape library)	S3 / Glacier	Replace physical tapes
Transfer Family	Online	Ongoing	Partners use SFTP / FTPS / FTP / AS2	S3, EFS	File exchange with partners
Snowball Edge	Offline (shipped device)	One-time	No	S3	Existing customers only (support ends Dec 31, 2026)
S3 Transfer Acceleration	Online (edge network)	Per upload	No	S3	Long-distance uploads
DMS	Online	Full load + CDC	No	Databases, S3, others	Databases with minimal downtime

Rule: One-time bulk copy → DataSync. On-premises apps keep using the share → Storage Gateway.

3. Database migration

Option	Engine change	Converts schema / code	Minimal downtime	Result
Native backup / restore	No	No	No (downtime during copy)	Simplest same-engine move
DMS	Either	No	Yes (full load + CDC)	Moves the data
SCT / DMS Schema Conversion	Yes	Yes (schema, stored procedures)	—	Run before DMS for a new engine
Babelfish for Aurora PostgreSQL	SQL Server → Aurora PostgreSQL	Understands T-SQL	—	Cuts SQL Server licences with few app changes
MGN of the database server	No	No	Yes	Database on EC2 — still self-managed

Rule: Different engine → SCT first, then DMS. "Managed database" rules out MGN.

4. Connecting VPCs

Option	Transitive routing	Overlapping CIDRs	Scale	Cost	Best for
VPC peering	No	No	1:1 (up to 125 active per VPC)	Data transfer only	A few VPCs
Transit Gateway	Yes	No	Thousands; regional hub, RAM-shared	Per attachment-hour + per GB	Many VPCs + on-premises
PrivateLink	— (one service, one way)	Yes	Many consumers	Per endpoint-hour + per GB	Expose one service
VPC sharing (RAM)	— (same VPC)	—	Many accounts in one VPC	No extra charge	Central network team owns the VPC
Cloud WAN	Yes	No	Global, many Regions	Core network + attachment fees	Global network from one policy
VPC Lattice	— (service level)	Yes	Many services and accounts	Per service + per request	Service-to-service with IAM auth

Rule: Overlapping CIDRs → PrivateLink or Lattice. Many VPCs needing full routing → Transit Gateway.

5. Links to on-premises

Option	Encrypted	Bandwidth	Setup time	Over the internet	Best for
Site-to-Site VPN	Yes (IPsec)	1.25 Gbps per tunnel (5 Gbps large tunnels on TGW / Cloud WAN); ECMP on TGW	Minutes	Yes	Quick start, backup for DX
Direct Connect — dedicated	No by default (MACsec on 10/100/400G, or VPN over DX)	1 / 10 / 100 / 400 Gbps	Weeks or more	No	Consistent, high bandwidth
Direct Connect — hosted	No by default	50 Mbps – 25 Gbps (partner)	Weeks (via partner)	No	Smaller dedicated capacity
Client VPN	Yes (TLS)	Per user	Minutes	Yes	Individual remote users

Rule: "Encrypted and consistent" → DX + VPN (or MACsec). "Start now" → VPN.

6. Front door options

Option	Layer / protocol	Static IP	Caching	AWS WAF	Notes
Application Load Balancer	Layer 7: HTTP / HTTPS, gRPC	No	No	Yes	Path/host rules, Cognito/OIDC auth, sticky sessions
Network Load Balancer	Layer 4: TCP / UDP / TLS	Yes (one Elastic IP per AZ)	No	No	Very high performance; PrivateLink provider
Gateway Load Balancer	Layer 3 (GENEVE)	No	No	No	Inline firewalls and appliances
CloudFront	HTTP / HTTPS at the edge	No (IPs change)	Yes	Yes	Global content, origin protection
Global Accelerator	TCP / UDP over anycast	Yes (2 static IPs)	No	No (put WAF on the ALB behind it)	Fast Regional failover, non-HTTP

Rule: UDP or static IPs → NLB / Global Accelerator. Caching → CloudFront.

7. Route 53 routing policies

Policy	Decides by	Health checks	Typical use
Simple	One record (one or more values)	No	Single resource
Weighted	Percentages you set	Yes	Gradual migration, A/B tests
Latency	Lowest measured latency to a Region	Yes	Send users to the fastest Region
Failover	Primary, else secondary	Yes (required on primary)	Active-passive DR
Geolocation	User's continent / country	Yes	Content rights, compliance, language
Geoproximity	Distance, with a bias to shift traffic	Yes	Move traffic between Regions
Multivalue answer	Up to 8 healthy records at random	Yes	Simple client-side spreading
IP-based	Client IP range (CIDR)	Yes	Route by network or ISP

Rule: Fastest Region → latency (not geolocation). DR → failover with health checks.

8. DR strategies

Strategy	RTO	RPO	Running in DR Region	Cost
Backup and restore	Hours	Hours	Nothing (backups only)	Lowest
Pilot light	Tens of minutes – hours	Minutes (data replicated)	Data replicated; compute off	Low
Warm standby	Minutes	Seconds – minutes	Scaled-down full copy	Medium
Multi-site active-active	Near zero	Near zero	Full capacity in 2+ Regions	Highest

Rule: Pick the cheapest row that meets both RTO and RPO.

9. Database high availability and DR

Option	Readable copy	Other Region	Failover	Data loss	Notes
RDS Multi-AZ DB instance	No	No	Automatic, ~1–2 min	None (synchronous)	HA only, not DR
RDS Multi-AZ DB cluster	Yes (2 standbys)	No	Automatic, typically < 35 s	None	MySQL / PostgreSQL
RDS read replica	Yes	Yes (cross-Region possible)	Manual promotion	Possible (asynchronous)	Read scaling, cheap DR
Aurora replica	Yes (up to 15)	No	Automatic, ~30 s	None (shared storage)	Read scaling + HA
Aurora global database	Yes (secondary Regions)	Yes (up to 10 secondary Regions)	Switchover (planned) / failover (outage); RTO < 1 min	RPO ~1 s (replication typically < 1 s)	Regional DR, local reads
DynamoDB global tables	Yes (read and write everywhere)	Yes	Active-active, nothing to promote	~1 s eventual, or zero with multi-Region strong consistency	Multi-Region NoSQL

Rule: Multi-AZ = one Region only. DR to another Region needs replicas, global DB or global tables.

10. Messaging and streaming

Service	Model	Ordering	Replay	Fan-out	Notes
SQS standard	Queue (consumers pull)	Best effort	No	No (one consumer per message)	Nearly unlimited throughput; DLQ
SQS FIFO	Queue	Yes (per message group)	No	No	Exactly-once processing, 5-min dedup
SNS	Pub/sub (push)	FIFO topics only	Partly (FIFO topic archive)	Yes	Push to SQS, Lambda, HTTP, email, SMS
EventBridge	Event bus with content rules	No	Yes (archive and replay)	Yes (rules → targets)	SaaS sources, schedules, cross-account
Kinesis Data Streams	Stream	Yes (per partition key)	Yes (24 h – 365 days)	Yes (several consumers, enhanced fan-out)	Real-time analytics
Amazon MSK	Kafka stream	Yes (per partition)	Yes	Yes	Existing Kafka workloads
Amazon MQ	Broker (ActiveMQ / RabbitMQ)	Per queue	No	Yes (topics)	JMS / AMQP / MQTT / STOMP lift-and-shift

Rule: Buffer → SQS. One event to many → SNS / EventBridge. Ordered replay by many readers → Kinesis.

11. Caching layers

Option	Where	What it caches	Use when
CloudFront	Edge locations	HTTP responses (static and dynamic)	Users far away; offload origin
API Gateway cache	REST API stage	API responses	Repeated GET calls (REST APIs only)
ElastiCache (Valkey / Redis OSS / Memcached)	Next to the app	Any data, sessions, leaderboards	Offload database reads
DAX	In front of DynamoDB	Items and queries (microseconds)	Read-heavy DynamoDB, no code rewrite
Read replicas	Database tier	Full copy of the data	Read scaling with SQL
MemoryDB	Primary database	Durable in-memory data	Redis-compatible store that must not lose data

Rule: Caches help reads, never writes.

12. Compute options

Option	Max runtime	Servers to manage	OS / host control	Scales to zero	Best for
Lambda	15 minutes	No	No	Yes	Short, event-driven code
Fargate (ECS / EKS)	No limit	No	No	Partly (tasks to 0)	Containers without servers
ECS / EKS on EC2	No limit	Yes (instances)	Yes	No	GPUs, custom AMIs, host agents
AWS Batch	No limit	No (managed environments)	Partly	Yes	Queued batch jobs (EC2, Spot, Fargate)
Elastic Beanstalk	No limit	Managed for you (EC2 in your account)	Partly	No	Web apps with little AWS knowledge
EC2	No limit	Yes	Full	No	Lift-and-shift, licences, special hardware
Step Functions	Standard: 1 year · Express: 5 min	No	No	Yes	Orchestration, retries, waits

Rule: Over 15 minutes → not Lambda. "Least overhead" → avoid managing EC2.

13. Storage options

Option	Type / protocol	Shared by many	Multi-AZ	OS	Best for
EBS	Block	No (Multi-Attach io1/io2, same AZ only)	No (one AZ)	Any	Boot volumes, databases
Instance store	Block, temporary	No	No	Any	Scratch, cache (lost on stop)
EFS	File (NFS)	Yes	Yes (Standard) / No (One Zone)	Linux	Shared Linux files
FSx for Windows File Server	File (SMB, AD, ACLs)	Yes	Multi-AZ option	Windows	Windows shares
FSx for Lustre	File (Lustre client)	Yes	No (one AZ)	Linux	HPC, ML, linked to S3
FSx for NetApp ONTAP	NFS + SMB + iSCSI	Yes	Multi-AZ option	Both	NetApp moves, multiprotocol
S3	Object (HTTP)	Yes	Yes (except One Zone classes)	Any	Data lakes, backups, static content

Rule: Pick by protocol first: NFS → EFS, SMB → FSx Windows, HPC → Lustre, both → ONTAP.

14. Deployment strategies

Strategy	Downtime	Extra cost	Capacity during deploy	Rollback
All at once	Yes	No	Drops to zero	Redeploy old version (slow)
Rolling	No	No	Reduced by one batch	Redeploy old version
Rolling with additional batch	No	Small	Full	Redeploy old version
Immutable	No	Double while deploying	Full	Terminate new instances (fast)
Blue/green	No	Double	Full	Switch traffic back (instant)
Canary / linear	No	Small	Full	Automatic on CloudWatch alarm

Rule: "No capacity loss" → rolling with batch, immutable or blue/green. "Small % first" → canary.

15. Policy types

Policy	Attached to	Can grant	What it limits
Identity-based	User, group, role	Yes	—
Resource-based	Bucket, key, queue, topic, function…	Yes (also cross-account)	—
Trust policy	Role	Yes (who may assume it)	—
Permissions boundary	User or role	No	Maximum for that identity
SCP	Root, OU, account	No	Maximum for principals in member accounts (not the management account)
RCP	Root, OU, account	No	Maximum access to resources in those accounts
Session policy	Passed when assuming a role	No	Maximum for that session

Rule: Explicit deny anywhere wins. Only identity, resource and trust policies grant.

16. Sign-in and directories

Option	For	Gives	Typical answer for
IAM users	People or apps	Long-term passwords / keys	Avoid (break-glass only)
IAM Identity Center	Employees	SSO to many accounts and apps, permission sets	Workforce access across accounts
IAM SAML / OIDC provider	Employees or CI/CD, per account	Temporary role credentials	Single-account federation, GitHub OIDC
Cognito user pools	App customers	Sign-up / sign-in, JWT, MFA, social login	Customer apps; ALB / API Gateway auth
Cognito identity pools	App users	Temporary AWS credentials	Users upload to S3 directly
AWS Managed Microsoft AD	Windows workloads	Real AD in AWS, trusts	RDS SQL Server, FSx, WorkSpaces
AD Connector	On-premises AD users	Proxy to on-premises AD (no copy)	WorkSpaces, Identity Center with on-prem AD
Simple AD	Small directories	Samba-based AD	Closed to new customers

Rule: Employees → Identity Center. Customers → Cognito.

17. Encryption options

Option	Who holds the key	Key use logged in CloudTrail	Cross-account	Notes
SSE-S3	AWS (S3)	No	—	Default encryption
SSE-KMS, AWS managed key	AWS, in your account	Yes	No	Can't edit its policy
SSE-KMS, customer managed key	You (policy, rotation)	Yes	Yes	S3 Bucket Keys cut KMS cost
SSE-C	You send the key with each request	No	—	HTTPS required
Client-side encryption	You, before upload	—	—	AWS Encryption SDK
CloudHSM	You (dedicated HSM)	Own logs	—	PKCS#11 / JCE; KMS custom key store
External key store (XKS)	You, outside AWS	Yes	—	Every call goes to your HSM

Rule: Must audit or share the key → customer managed KMS key.

18. Security services

Service	Does	Prevents	Typical answer for
CloudTrail	Records API calls	No	Who did what, when
AWS Config	Records configuration, checks rules	No (can auto-fix via SSM)	Compliance history, drift
GuardDuty	Threat detection from logs	No	Compromised credentials, crypto-mining
Inspector	Vulnerability scans (EC2, ECR, Lambda)	No	CVEs
Macie	Finds sensitive data in S3	No	PII discovery
Detective	Investigates findings	No	Root cause of a finding
Security Hub	Aggregates findings, checks standards	No	One security view
IAM Access Analyzer	Finds external / unused access, generates policies	No	Resources shared outside the org
AWS WAF	Filters web requests	Yes	SQL injection, bots, rate limits
Shield Advanced	DDoS protection + response team	Yes	Large DDoS, cost protection
Network Firewall	Filters VPC traffic	Yes	Egress filtering, IPS
Firewall Manager	Applies WAF / firewall / SG rules org-wide	Yes (deploys rules)	Rules on all accounts, incl. new

Rule: Detect → the "No" rows. Block → WAF, Shield, Network Firewall, SCP.

19. Secrets and keys

Service	Stores	Automatic rotation	Cross-account	Cost
Secrets Manager	Passwords, API keys	Yes (built-in)	Yes (resource policy)	Per secret per month
Parameter Store	Configuration, SecureString values	No	Advanced tier (via RAM)	Standard tier free
KMS	Encryption keys (not secrets)	Yes (key material: scheduled or on demand)	Yes (key policy)	Per key + per request

Rule: "Rotate passwords" → Secrets Manager.

20. Purchase options

Option	Discount	Flexibility	Covers Fargate / Lambda	Commitment
On-Demand	None	Full	Yes	None
Compute Savings Plan	Up to 66%	Any family, size, Region, OS	Yes	1 or 3 years, $/hour
EC2 Instance Savings Plan	Up to 72%	One family in one Region	No	1 or 3 years, $/hour
Standard Reserved Instance	Up to 72%	Fixed type (size flexible within family)	No	1 or 3 years
Convertible Reserved Instance	Up to 66%	Can exchange for other types	No	1 or 3 years
Spot	Up to 90%	Can be interrupted (2-min notice)	Partly (Fargate Spot)	None
Database Savings Plan	Up to 35% serverless, 20% provisioned	Across database services and engines	—	1 year
Dedicated Host	—	Whole server; socket/core visibility for BYOL	No	On-Demand or reservation
Capacity Reservation	None by itself	Guaranteed capacity in one AZ	No	Billed whether used or not

Rule: Workload will change → Compute Savings Plan. Interruptible → Spot.

21. S3 storage classes

Class	Retrieval time	Min. storage duration	Retrieval fee	AZs	Use for
Standard	Milliseconds	None	No	3+	Frequently accessed data
Intelligent-Tiering	Milliseconds (optional archive tiers slower)	None	No (small monitoring fee per object)	3+	Unknown or changing access
Standard-IA	Milliseconds	30 days (min. 128 KB billed)	Yes (per GB)	3+	Infrequent but instant
One Zone-IA	Milliseconds	30 days (min. 128 KB billed)	Yes (per GB)	1	Re-creatable infrequent data
Glacier Instant Retrieval	Milliseconds	90 days (min. 128 KB billed)	Yes (per GB)	3+	Archive read about once a quarter
Glacier Flexible Retrieval	Expedited 1–5 min · Standard 3–5 h · Bulk 5–12 h (free)	90 days	Yes (bulk free)	3+	Archive, minutes to hours is fine
Glacier Deep Archive	Standard within 12 h · Bulk within 48 h	180 days	Yes	3+	Long-term compliance archive

Rule: Unknown access → Intelligent-Tiering. Hours are acceptable → Flexible / Deep Archive. Re-creatable → One Zone-IA.

22. EBS volume types

Type	Max IOPS	Max throughput	Size	Boot volume	Use for
gp3 (SSD)	80,000	2,000 MB/s	1 GB – 64 TB	Yes	Default; IOPS and throughput set independently of size
gp2 (SSD)	16,000	250 MB/s	1 GB – 16 TB	Yes	Older default; IOPS grow with size
io2 Block Express (SSD)	256,000	4,000 MB/s	4 GB – 64 TB	Yes	Highest performance databases; Multi-Attach
io1 (SSD)	64,000	1,000 MB/s	4 GB – 16 TB	Yes	Provisioned IOPS (older); Multi-Attach
st1 (HDD)	500	500 MB/s	125 GB – 16 TB	No	Big sequential reads: logs, data warehouses
sc1 (HDD)	250	250 MB/s	125 GB – 16 TB	No	Cold, rarely accessed data, lowest cost

Rule: Predictable IOPS cheaply → gp3. Highest IOPS / Multi-Attach → io2. Throughput-heavy sequential → st1.

23. Analytics services

Service	Serverless	Data stays in S3	Query / work type	Use for
Athena	Yes	Yes	SQL, pay per data scanned	Ad-hoc SQL on S3 (Parquet + partitions cut cost)
Redshift (provisioned / Serverless)	Partly (Serverless option)	No (loaded into Redshift)	Data warehouse SQL	Complex, frequent BI queries
Redshift Spectrum	No (needs a Redshift cluster)	Yes	SQL joining Redshift + S3	Warehouse queries that include S3 data
EMR	Partly (EMR Serverless)	Yes (or HDFS)	Spark, Hive, Presto, Hadoop	Big data processing with open-source tools
Glue	Yes	Yes	ETL jobs, crawlers, Data Catalog	Transform data; catalog for Athena/Redshift
QuickSight	Yes	—	Dashboards (SPICE or direct query)	Business dashboards

Rule: Occasional SQL on S3 → Athena. Frequent heavy BI → Redshift. Spark/Hadoop → EMR. ETL → Glue.

24. Streaming family

Service	Latency	Replay	Consumers	Destinations	Use for
Kinesis Data Streams	Real time (sub-second)	Yes (24 h – 365 days)	Many (custom apps, Lambda, enhanced fan-out)	Your consumers	Real-time processing, several readers
Amazon Data Firehose	Near real time (buffered)	No	Delivery only	S3, Redshift, OpenSearch, Splunk, HTTP	Load streaming data with no code
Managed Service for Apache Flink	Real time	—	Reads from KDS or MSK	KDS, Firehose, S3, others	Stream analytics (SQL / Java / Python)
Amazon MSK	Real time	Yes	Many (Kafka consumers)	Your consumers, MSK Connect	Existing Kafka

Rule: Firehose is a sink, never a source for Flink. Replay or multiple consumers → Kinesis Data Streams.

25. In-memory options

Option	Replication / Multi-AZ	Durable (no data loss)	Data types	Use for
ElastiCache (Valkey / Redis OSS)	Yes	No (cache; backups possible)	Rich (lists, sets, sorted sets…)	Cache, sessions, leaderboards
ElastiCache (Memcached)	No	No	Simple key-value, multithreaded	Simple cache that can be lost
MemoryDB	Yes	Yes (Multi-AZ transaction log)	Redis-compatible	Primary database needing microsecond reads
DAX	Yes (cluster across AZs)	No (write-through to DynamoDB)	DynamoDB items/queries	DynamoDB reads in microseconds, no code rewrite

Rule: Must not lose data → MemoryDB. DynamoDB reads → DAX. Sessions with replication → ElastiCache Valkey/Redis OSS.

26. Lock modes (S3 Object Lock and AWS Backup Vault Lock)

Mode	Who can remove or shorten	Root user can bypass	Use for
Object Lock — governance	Users with special permission (s3:BypassGovernanceRetention)	With that permission	Protect against most users; testing settings
Object Lock — compliance	Nobody until retention ends	No	Regulatory WORM
Object Lock — legal hold	Users with s3:PutObjectLegalHold; no end date	With that permission	Hold during litigation
Vault Lock — governance	Users with the right IAM permissions	With permission	Protect backups, still adjustable
Vault Lock — compliance	Nobody after the grace time ends	No	Ransomware-proof backups

Rule: "Even root / nobody can delete" → compliance mode.

27. Firewalls and filters

Option	Where it applies	Stateful	Allow / deny	Filters by	Use for
Security group	Instance / ENI	Yes	Allow only	IP, port, other SGs	Default per-workload control
Network ACL	Subnet	No (rules both ways)	Allow and deny	IP, port	Block specific IPs at subnet level
AWS Network Firewall	VPC (routes through firewall endpoints)	Yes (and stateless rules)	Both	IP, port, domain, IPS signatures	Central egress / east-west inspection
AWS WAF	CloudFront, ALB, API Gateway, AppSync, Cognito	—	Both	HTTP content, IP, geo, rate	Web exploits, bots, rate limits
Route 53 Resolver DNS Firewall	VPC DNS queries	—	Both	Domain lists	Block or allow domains looked up from VPCs

Rule: Deny one IP → NACL (SGs can't deny). Domain filtering → Network Firewall or DNS Firewall.

28. VPC endpoint types

Type	Services	How it works	Cost	Reachable from on-premises
Gateway endpoint	S3, DynamoDB	Route table entry	Free	No
Interface endpoint (PrivateLink)	Most AWS services, endpoint services	ENI with private IP, security group, private DNS	Per hour + per GB	Yes (over DX / VPN)
Gateway Load Balancer endpoint	Your inspection appliances	Route table target to a GWLB	Per hour + per GB	—

Rule: Cheap S3 from a VPC → gateway endpoint. S3 privately from on-premises → interface endpoint.

29. Edge code

Option	Runs at	Max run time	Network / body access	Language	Use for
CloudFront Functions	Viewer request / response	Sub-millisecond	No	JavaScript	Header changes, redirects, cache-key normalization, simple auth
Lambda@Edge	Viewer and origin request / response	Up to 30 s	Yes	Node.js, Python	Heavier logic, call other services, read the body

Rule: Normalize cache keys or simple redirects → CloudFront Functions. Needs network or the body → Lambda@Edge.

30. API Gateway types

Type	Usage plans / API keys	Caching	Private endpoint	AWS WAF	Auth	Use for
REST API	Yes	Yes	Yes	Yes	IAM, Cognito, Lambda authorizer	Full-featured APIs, per-customer quotas
HTTP API	No	No	No (Regional only)	No	IAM, JWT (incl. Cognito), Lambda authorizer	Cheaper, simpler proxy APIs
WebSocket API	API key on $connect	No	No	No	IAM, Lambda authorizer	Two-way real-time messages

Rule: Usage plans, caching, private or WAF → REST API.

31. Who records what

Service	Records	Answers the question	Use for
CloudWatch metrics / alarms	Numbers over time	How is it performing?	Alerts, scaling
CloudWatch Logs	Application and service logs	What did the app log?	Troubleshooting, subscriptions
CloudTrail	API calls	Who did what, when?	Audit, security
AWS Config	Resource configuration over time	What changed? Is it compliant?	Compliance, drift
X-Ray	Request traces	Where is the latency or error?	Distributed tracing
CloudWatch Application Signals	Service health, SLOs	Are we meeting our targets?	Error budgets, burn rate
CloudWatch Synthetics	Scripted canary tests	Does the flow work right now?	Test when real users are few
CloudWatch RUM	Real user experience in browsers	What do users actually see?	Front-end performance

Rule: "Who deleted it" → CloudTrail. "What did it look like before" → Config.

32. Route 53 ARC features

Feature	Trigger	Scope	Use for
Routing controls	Manual (on/off switch, highly available data plane)	Region or cell	Manual Regional failover with safety rules
Zonal shift	Manual, temporary	One AZ	Move traffic away from an impaired AZ
Zonal autoshift	Automatic by AWS (weekly practice runs)	One AZ	AWS shifts traffic when it detects AZ impairment
Region switch	Plan you execute	Multi-Region application	Orchestrated, auditable failover and failback
Readiness checks	—	—	No longer open to new customers (use Region switch)

Rule: Manual Region switch → routing controls. Bad AZ → zonal shift / autoshift. Whole runbook → Region switch.

33. Control Tower controls

Type	How it works	Implemented with	Example
Preventive	Blocks the action	SCPs / RCPs (and declarative policies)	Deny actions outside approved Regions
Detective	Finds non-compliant resources	AWS Config rules	Detect unencrypted RDS instances
Proactive	Checks resources before CloudFormation creates them	CloudFormation Hooks	Block a non-compliant resource in a stack

Rule: Guidance levels: mandatory, strongly recommended, elective. Proactive only checks resources created through CloudFormation.

34. Step Functions workflow types

Type	Max duration	Execution semantics	Callback (task token) / .sync	Priced by	Use for
Standard	1 year	Exactly-once	Yes	State transitions	Long, auditable workflows, human approvals, payments
Express (asynchronous)	5 minutes	At-least-once	No	Executions, duration, memory	High-volume, idempotent event processing
Express (synchronous)	5 minutes	At-most-once	No	Executions, duration, memory	Orchestrate microservices behind an API

Rule: Waits for a human or runs long → Standard.

35. Cost tools

Tool	Does	Typical answer for
Cost Explorer	Views, filters and forecasts spend; RI/SP recommendations	Analyze past cost and trends
AWS Budgets	Alerts on actual or forecast spend; budget actions	Alert or act at a threshold
CUR 2.0 (Data Exports)	Hourly, resource-level billing data to S3	Detailed chargeback with Athena
Cost Anomaly Detection	ML alerts on unusual spend	Spikes with no fixed threshold
Compute Optimizer	Rightsizing recommendations	Over-provisioned EC2, EBS, Lambda, ECS, RDS
Trusted Advisor	Best-practice checks (cost, security, limits…)	Idle resources, open security groups
Cost Optimization Hub	One deduplicated list of savings recommendations	Prioritize savings across the organization
Billing Conductor	Custom pricing and pro forma bills	Rebill subsidiaries with a markup

Rule: Alert → Budgets. Unusual spike → Anomaly Detection. Rightsize → Compute Optimizer.

36. Running outside a Region

Option	Hardware owner	Location	Needs Region connection	Use for
Outposts (racks / servers)	AWS (installed on site)	Your data center	Yes (service link)	Low latency or local data with AWS services on premises
Local Zones	AWS	AWS site in a metro area	Part of a parent Region	Single-digit ms latency to a city
Wavelength Zones	AWS (in carrier 5G networks)	Telecom provider sites	Part of a parent Region	Mobile / 5G edge apps
ECS Anywhere	You	Your servers	Yes (outbound)	Same ECS tools on your own servers
EKS Anywhere	You	Your servers (e.g. vSphere)	No (can run air-gapped)	EKS-compatible Kubernetes on premises
Snowball Edge compute	AWS device	Remote / disconnected site	No	Edge compute; no longer available to new customers

Rule: Disconnected site → EKS Anywhere / Snowball Edge. AWS services in your data center → Outposts.